

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-1999

Tulloch 138 kV
October 02, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-1999. The Interconnection Customer proposes a 136.6 MW natural gas combined cycle facility interconnecting at the Tulloch 138 kV substation in the Pennsylvania area. The request was received on February 06, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 554 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,746,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

GRANADA - KINDER MORGAN SUBSTATION 138 KV: 115.0% loading (192 MVA) following the outage of the TOMBSTONE - BIXBY HILL 500 KV; DFAX = 6.5%.

The ROBBIE - KINDE MORGAN 138 KV is overloaded to 106.3% of its 205 MVA emergency rating for the loss of the BIXBY HILL - TOMBSTONE 500 KV. The project contributes 6.5% DFAX (8.9 MW) to this overload.

The WHITE HEATH - TULLOCH SUBSTATION 138 KV is overloaded to 100.1% of its 211 MVA emergency rating for the loss of the MERCURY - ELK RIV 345 KV line. The project contributes 27.7% DFAX (37.8 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
KINDER MORGAN - GRANADA 138 KV	Rebuild line with higher-capacity conductor	\$2,172,000	100% (\$2,172,000)
ROBBIE - KINDE MORGAN 138 KV	Reconductor line	\$2,137,000	100% (\$2,137,000)
WHITE HEATH - TULLOCH SUBSTATION 138 KV	Reconductor line	\$943,000	46% (\$437,000)
	Total allocated to project		\$4,746,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.